Software Requirement Specification
for

HaChar ManaRes

System Analysis & Design

CSE 3412

Submitted
to

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Submitted
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Preliminary Sections

Abstract

HaChar ManaRes is a digital platform for Bangladesh's Haor/Char regions integrating disaster response with community management. Core features: real-time flood monitoring, boat rescue coordination, relief distribution, early warnings, agricultural support, and NGO coordination. Research shows 77.8% face communication breakdown during floods; 74.1% support digital solutions. Built as PWA with offline capabilities, the system targets 40-50% faster response and 60-70% improved relief transparency.

Acknowledgement

We thank **Rizvan Jawad Ruhan Sir** for guidance; **30 community respondents** for insights; **United International University** for resources; and **Team SAD** members: Mir Mohammed Mohiuddin, Md. Sabih Sadman, Koishik Ahmed, Ibnul Tahsin, Robayet Rowshan Siam.

Executive Summary

Problem: 77.8% cite communication breakdown during floods; only 14.8% satisfied with local management; corruption in relief distribution. 88.9% have smartphones, 63% internet access - strong tech readiness.

Solution Features

Disaster Response:	Flood monitoring, boat rescue, early warnings, evacuation
Community Support:	Crop loss reporting, fishing permits, livestock health
Management:	Relief tracking, NGO coordination, volunteer tasks
Safety:	School attendance tracking, shelter management

Technical Stack

React.js + Node.js + MongoDB + Leaflet maps, Bangla ui, voice for illiterate users.

Implementation

Pilot (6mo): 3-5 communities (5k residents) → **Expansion (2yr):** 20+ communities → **Sustainability:** Community-led maintenance.

Impact Metrics

Response time reduction:	40-50%
Relief transparency:	60-70% improvement
Community participation:	30-40% increase
Tech adoption:	74.1% community support

Chapter 1

Introduction & Project Proposal

1.1 Project Summary and Proposal

The **HaChar ManaRes System** (Haor/Char Community Management and Disaster Response System) is a comprehensive digital platform designed to address the unique challenges faced by vulnerable Haor and Char regions in Bangladesh. This system integrates disaster response, community management, and resource coordination into a unified platform that connects residents, local authorities, NGOs, volunteers, farmers, and fishermen. By leveraging real-time information and digital tools, the system aims to improve disaster preparedness, ensure timely relief distribution, and support sustainable community development throughout the year.

1.2 Motivation & Problem Statement

The Haor and Char regions of Bangladesh face recurrent floods, cyclones, and other natural disasters that severely impact livelihoods, particularly for farmers and fishermen. Current disaster management systems suffer from:

- **Inefficient communication:** 77.8% of surveyed residents cite communication as the biggest challenge during floods
- **Limited transparency:** Corruption and mismanagement in relief distribution systems
- **Delayed response:** Lack of real-time information about water levels and rescue resources
- **Fragmented coordination:** Disconnect between authorities, NGOs, and community members
- **Information gaps:** Inadequate tracking of vulnerable populations and their needs

The **HaChar ManaRes System** addresses these challenges by providing a unified digital platform for coordinated disaster response and community management.

1.3 Goals & Objectives

- **Primary Goal:** Enhance disaster resilience and community management in Haor/Char regions through digital innovation
- **Specific Objectives:**
  1. Establish real-time communication channels for emergency alerts and updates
  2. Improve transparency in relief distribution and resource allocation
  3. Enable efficient rescue operations through boat transport coordination
  4. Support agricultural and fishing communities with damage reporting and compensation tracking
  5. Facilitate data-driven decision making for authorities and NGOs
  6. Ensure inclusive access for all community members, including illiterate users
  7. Provide comprehensive community management tools beyond disaster response

1.4 Scope & Boundaries

1.4.1 Included Features (Prototype Scope):

- Resident registration and verification system
- Real-time flood risk monitoring and water level updates
- Boat transport request and management system
- Relief stock and distribution management
- Early warning and weather alert system
- Farmer crop loss reporting and verification
- Fishing permit and water body management
- Emergency evacuation planning and shelter management
- NGO activity coordination
- Analytics dashboard with real-time visualization

1.4.2 Excluded Features (for this prototype phase):

- Full-scale financial transaction processing for microloans
- Advanced AI-based disaster prediction models
- Integration with national meteorological satellite systems
- Offline synchronization for prolonged internet outages
- Multi-language support beyond Bangla and English
- Advanced GIS mapping with 3D visualization

Scale: The system will initially target 5-10 Haor/Char communities with approximately 10,000-20,000 residents, with scalability built for regional expansion.

1.5 System Development Life Cycle (SDLC)

The project follows a structured **Agile/Scrum methodology** with clear phases to ensure systematic development and delivery of MindTrack. The SDLC process is designed to be iterative, allowing for continuous improvement and adaptation to user feedback throughout the 1.5-month development timeline.

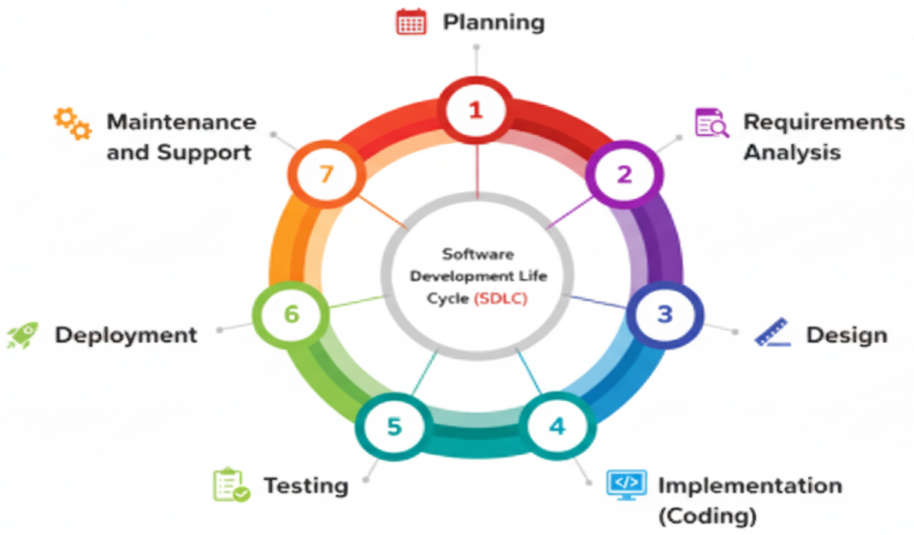


Figure 1.1: HaChar ManaRes System Development Life Cycle

1.5.1 SDLC Phases

1. Planning

- **Objective:** Establish clear project goals and understand user requirements
- **Activities:**
  - Define project scope and objectives
  - Identify target user demographics (students and professionals)
  - Conduct market research and competitive analysis
  - Establish success metrics and key performance indicators (KPIs)
- **Deliverables:** Project charter, scope document, initial risk assessment

2. Requirements Gathering

- **Objective:** Capture detailed functional and non-functional requirements
- **Activities:**
  - Conduct user surveys and interviews (29 respondents)
  - Analyze competitor applications (Calm, Headspace, Daylio, Moodfit, Finch)
  - Document user stories and use cases
  - Define technical specifications and constraints
- **Deliverables:** Software Requirements Specification (SRS), user stories, use case diagrams

3. System Design

- **Objective:** Create architectural and detailed design specifications
- **Activities:**
  - Design system architecture and database schema
  - Create UI/UX wireframes and prototypes
  - Define API specifications and integration points
  - Plan technical infrastructure and deployment strategy

- **Deliverables:** System architecture diagrams, database design, UI mockups, technical design document

4. Implementation (Building the Features)

- **Objective:** Implement core features and functionalities
- **Activities:**
  - Develop mobile application using Flutter/React Native
  - Implement backend services using Firebase
  - Integrate AI APIs for personalized recommendations
  - Build core modules: mood tracking, habit tracker, AI chatbot, productivity tools
- **Deliverables:** Functional application code, integrated modules, API endpoints

5. Testing

- **Objective:** Ensure system quality, reliability, and performance
- **Activities:**
  - Conduct unit testing, integration testing, and system testing
  - Perform user acceptance testing (UAT) with selected users
  - Test security features and data privacy measures
  - Validate performance under different load conditions
- **Deliverables:** Test cases, bug reports, test summary report, UAT sign-off

6. Deployment

- **Objective:** Launch the platform for user access
- **Activities:**
  - Deploy application to app stores (Google Play Store initially)
  - Configure production environment and monitoring tools
  - Conduct final security and performance checks
  - Prepare user documentation and support materials
- **Deliverables:** Live application, deployment documentation, user guides

7. Maintenance and Updates

- **Objective:** Ensure continuous improvement and user satisfaction
- **Activities:**
  - Monitor system performance and user feedback
  - Address bugs and technical issues promptly
  - Implement feature enhancements based on user requests
  - Plan and execute regular updates and improvements
- **Deliverables:** Update releases, performance reports, user feedback analysis

1.5.2 Timeline and Milestones

Sprint	Duration	Key Deliverables
Planning	Week 1	Project charter, scope definition
Requirements	Week 1-2	SRS document, user stories
Design	Week 2	System architecture, UI prototypes
Implementation	Week 3-5	Core features implementation
Testing	Week 5-6	Quality assurance
Deployment	Week 6	Deployment
Maintenance	Ongoing	Updates and improvements

Table 1.1: Project Timeline and Milestones

This structured SDLC approach ensures that MindTrack not only meets user expectations but also maintains high quality standards while allowing for flexibility and adaptation throughout the development process. The iterative nature of Agile methodology enables the team to respond quickly to changing requirements and incorporate user feedback effectively.

Chapter 2

Information Gathering & System Study

2.1 Introduction to Information Gathering

The information gathering phase focused on understanding the specific needs, challenges, and technological readiness of Haor/Char communities. A mixed-methods approach was employed, combining quantitative surveys with qualitative interviews to ensure comprehensive data collection from all stakeholder groups.

2.2 Information Sources

2.2.1 Internal Sources

- Bangladesh Water Development Board (BWDB) historical data
- Department of Disaster Management (DDM) reports and protocols
- Local NGO field reports and assessments

2.2.2 External Sources

- Academic research on flood management in Bangladesh
- International disaster response frameworks (UN OCHA, IFRC)
- Global best practices in community-based disaster management
- Technology solutions for rural and remote area connectivity

2.3 Information Gathering Methods

The following methods were employed to gather relevant information:

Literature Review: Analysis of existing research on Haor/Char ecosystems, flood management strategies, and digital solutions for rural communities.

Community Surveys: Structured questionnaire administered to 27 residents from Haor/Char regions, focusing on:

- Current disaster response experiences

- Communication preferences and technology access
- Priority needs during floods
- Willingness to adopt digital solutions

2.4 Define and Desired State

2.4.1 Current State (Define)

- **Communication:** Fragmented, unreliable during disasters, heavy reliance on word-of-mouth
- **Response Coordination:** Manual, delayed, limited visibility of resources and needs
- **Information Flow:** One-way from authorities, limited community feedback mechanisms
- **Resource Management:** Paper-based records, prone to errors and manipulation
- **Community Engagement:** Passive recipients rather than active participants
- **Data Utilization:** Limited analysis and reporting capabilities

2.4.2 Desired State

- **Communication:** Multi-channel, real-time, bidirectional information flow
- **Response Coordination:** Integrated platform connecting all stakeholders
- **Information Flow:** Transparent, accessible, inclusive of all community members
- **Resource Management:** Digital tracking with audit trails and transparency reports
- **Community Engagement:** Active participation in disaster preparedness and response
- **Data Utilization:** Analytics-driven decision making and predictive capabilities

Chapter 3

System Analysis & Evaluation

3.1 Introduction to System Analysis

This chapter gives a thorough look at the information gathered, with a focus on comprehending the market, assessing current solutions, finding important holes, and judging HaChar ManaRes's general viability and strategic positioning. This research will be used to figure out what the system needs and how it should be built.

3.2 Benchmark & Comparative Analysis

This section provides a detailed comparison of HaChar ManaRes with its key competitors, highlighting their strengths, weaknesses, and how MindTrack differentiates itself.

Feature	HaChar ManaRes	Sahana EDEN	Ushahidi	Disaster AWARE	Esri / ArcGIS	Vassar Labs (DIMS)
Resident registration	✅ Yes	✅ Partial	❌ No	❌ No	❌ No	❌ No
Flood risk map	✅ Yes	✅ Partial	✅ Partial	✅ Yes	✅ Yes	✅ Yes
Boat transport request	✅ Yes	❌ No	❌ No	❌ No	❌ No	❌ No
Relief stock management	✅ Yes	✅ Yes	❌ No	❌ No	✅ Partial	❌ No
Water level monitoring (manual)	✅ Yes	✅ Partial	✅ Partial	✅ Yes	✅ Partial	✅ Yes
Local school attendance tracking	✅ Yes	❌ No	❌ No	❌ No	❌ No	❌ No
Farmer crop loss reporting	✅ Yes	✅ Partial	✅ Partial	❌ No	✅ Partial	✅ Partial
Fishing permit management	✅ Yes	❌ No	❌ No	❌ No	❌ No	❌ No
Livestock vaccination log	✅ Yes	✅ Partial	❌ No	❌ No	❌ No	❌ No
Homestead protection suggestions	✅ Yes	❌ No	❌ No	✅ Partial	✅ Partial	✅ Partial
NGO assistance tracker	✅ Yes	✅ Yes	❌ No	❌ No	✅ Partial	✅ Partial
Microloan management	✅ Yes	❌ No	❌ No	❌ No	❌ No	❌ No
Emergency evacuation planning	✅ Yes	✅ Partial	✅ Partial	✅ Partial	✅ Yes	✅ Partial
Weather alert	✅ Yes	❌ No	✅ Partial	✅ Yes	✅ Partial	✅ Yes
Community meeting scheduling	✅ Yes	✅ Partial	❌ No	❌ No	❌ No	❌ No

Figure 3.1: Benchmark Analysis

3.3 Gap Analysis

Current Capabilities	Community Needs	Gaps
Manual record keeping	Digital tracking	Lack of automated systems
SMS alerts only	Multi-channel notifications	Limited communication options
Isolated systems	Integrated coordination	Fragmented stakeholder communication
Reactive response	Proactive prevention	Limited predictive capabilities
Centralized decision making	Community participation	Low community engagement
Opaque processes	Transparent operations	Limited visibility and accountability

Table 3.1: Gap Analysis: Current vs. Required System

3.4 Feasibility Analysis

The practicality of HaChar ManaRes has been assessed across technical, operational, and economic dimensions.

3.4.1 Technical Feasibility:

- **Technology Availability:** 88.9% smartphone usage and 63% internet access support digital implementation
- **Infrastructure Requirements:** Progressive Web App (PWA) approach minimizes device and connectivity constraints
- **Integration Capabilities:** APIs available for weather data, SMS services, and government databases
- **Security Requirements:** Standard encryption and authentication protocols can be implemented

3.4.2 Operational Feasibility:

- **User Acceptance:** 74.1% community support for digital platform
- **Training Requirements:** Simplified interface with Bangla support reduces training needs
- **Organizational Impact:** Aligns with government digital Bangladesh initiatives
- **Maintainability:** Modular design allows for incremental updates and maintenance

3.4.3 Economic Feasibility:

- **Development Costs:** Moderate, using open-source technologies and existing infrastructure
- **Operational Costs:** Low, with cloud hosting and automated processes

- **Return on Investment:** High, through reduced disaster losses and improved resource allocation
- **Funding Sources:** Potential support from government, NGOs, and development partners

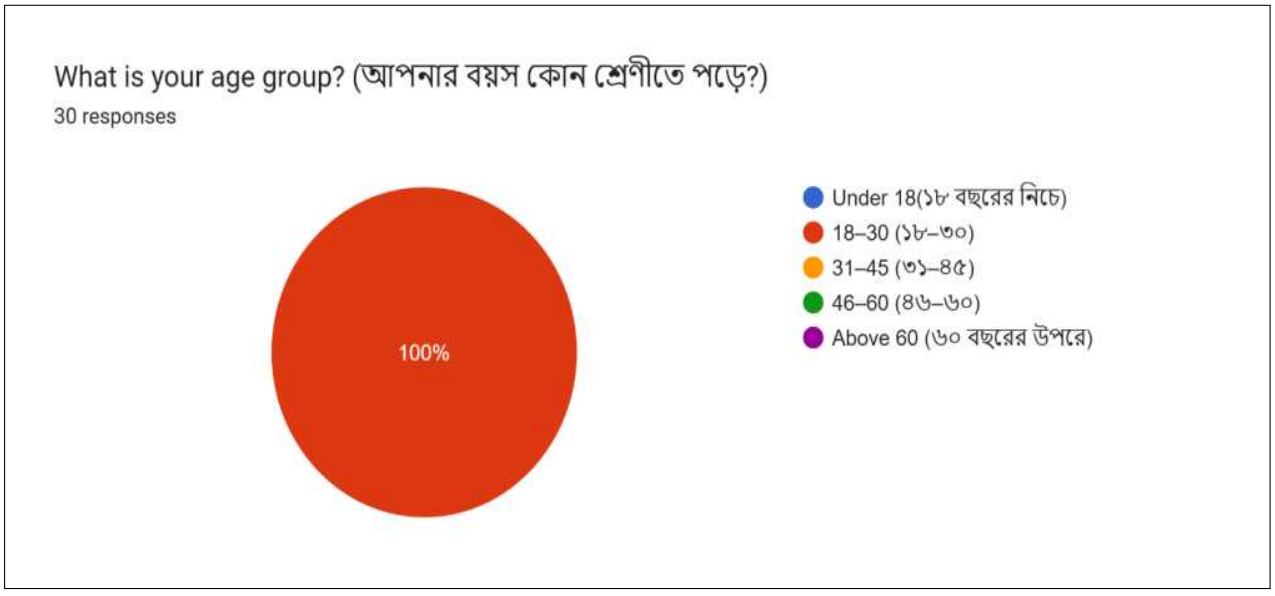
3.5 Survey

In the pursuit of continuous improvement and customer-centric product development, we conducted a Customer Satisfaction Survey to gauge the sentiment and feedback of our valued customers regarding our latest product. The survey aimed to identify areas for improvement, and understand how our product will align with customer expectations.

3.5.1 Survey Methodology

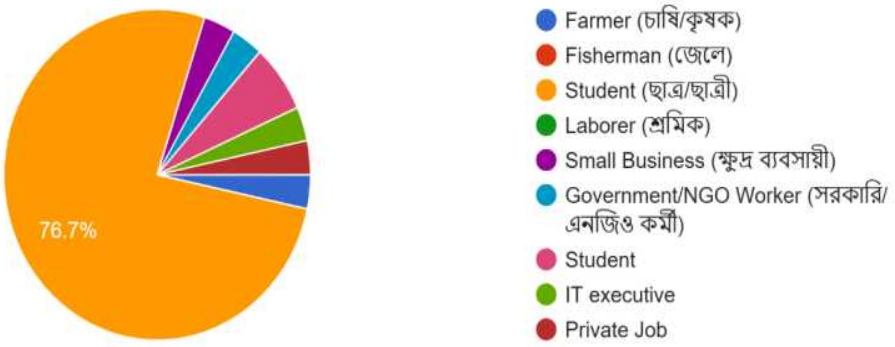
- Survey Type: Online questionnaire.
- Sample Size: 30 respondents.

3.6 Survey Results & Findings



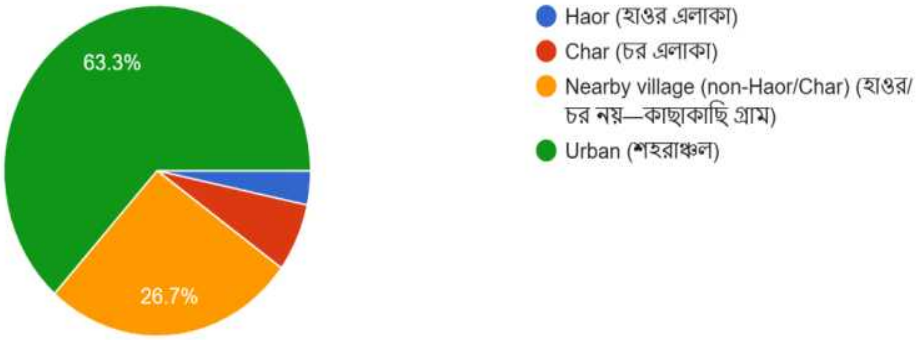
What is your occupation? (আপনার পেশা কী?)

30 responses



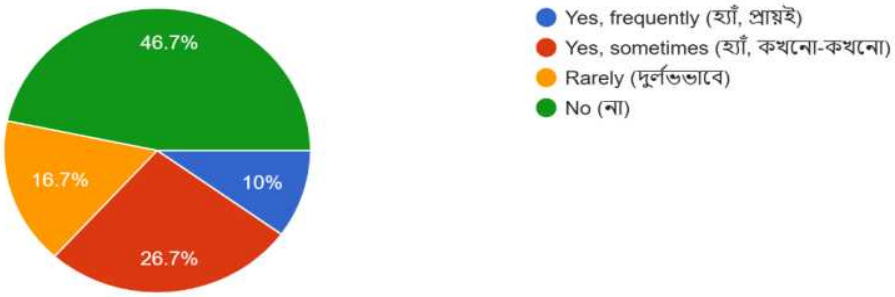
What type of area do you live in? (আপনি কোন ধরনের এলাকায় বসবাস করেন?)

30 responses



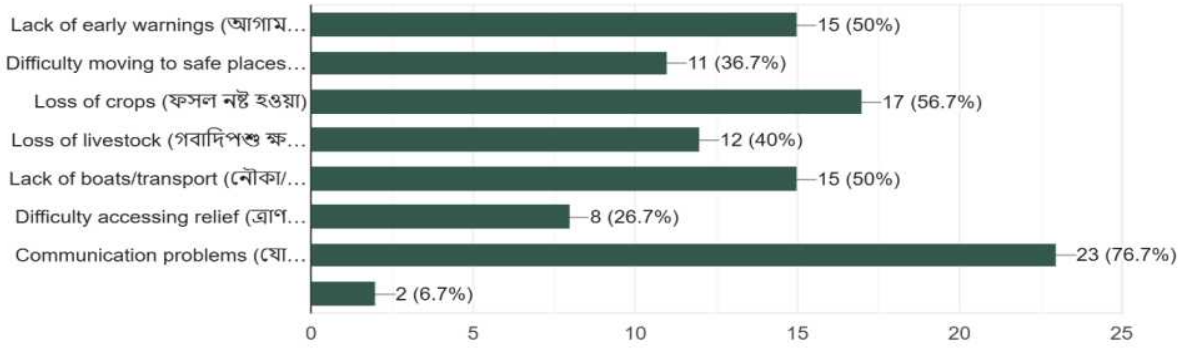
Have you or your family faced flood-related problems in the last 5 years? (গত ৫ বছরে আপনি বা আপনার পরিবার কি বন্যা-সংক্রান্ত সমস্যায় পড়েছেন?)

30 responses



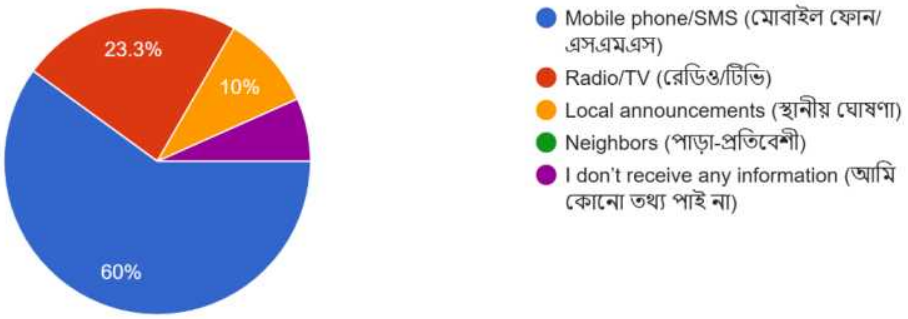
Which challenges do you face during floods? (বন্যার সময় আপনি কোন সমস্যাগুলোতে পড়েন?)

30 responses



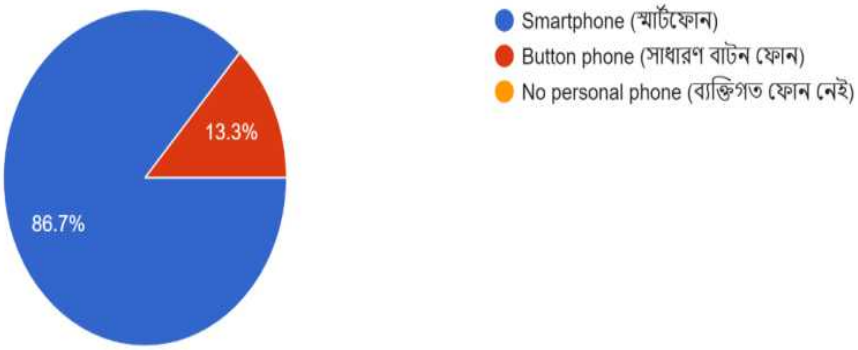
How do you currently receive flood or weather information? (আপনি বর্তমানে কীভাবে বন্যা বা আবহাওয়ার তথ্য পান?)

30 responses



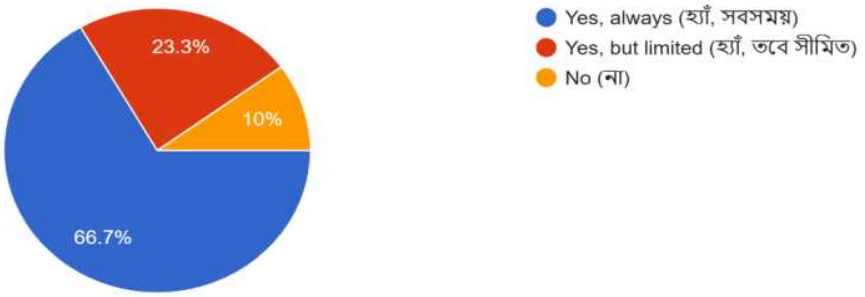
What type of mobile phone do you use? (আপনি কোন ধরনের মোবাইল ফোন ব্যবহার করেন?)

30 responses



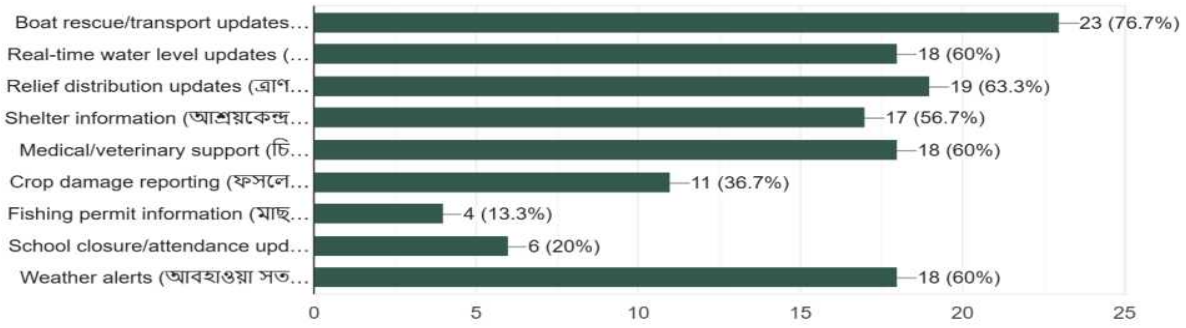
Do you have regular internet access on your phone? (আপনার মোবাইলে কি নিয়মিত ইন্টারনেট ব্যবহারের সুবিধা আছে?)

30 responses



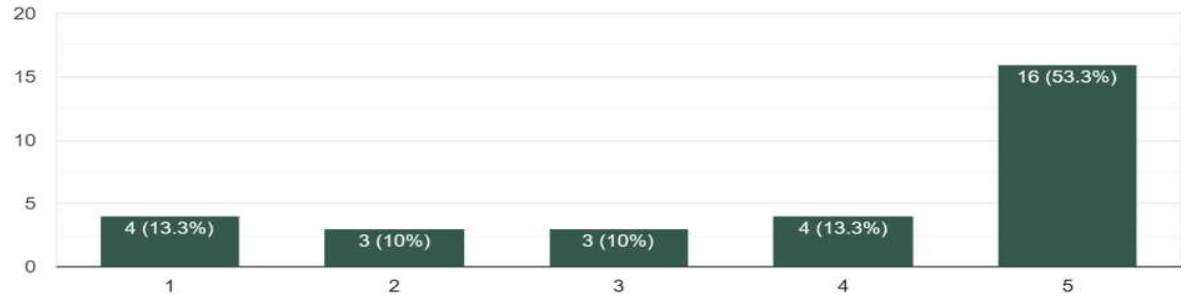
Which services would be most helpful for your community during floods? (বন্যার সময় আপনার কমিউনিটির জন্য কোন সেবাগুলো সবচেয়ে বেশি উপকারী হবে?)

30 responses



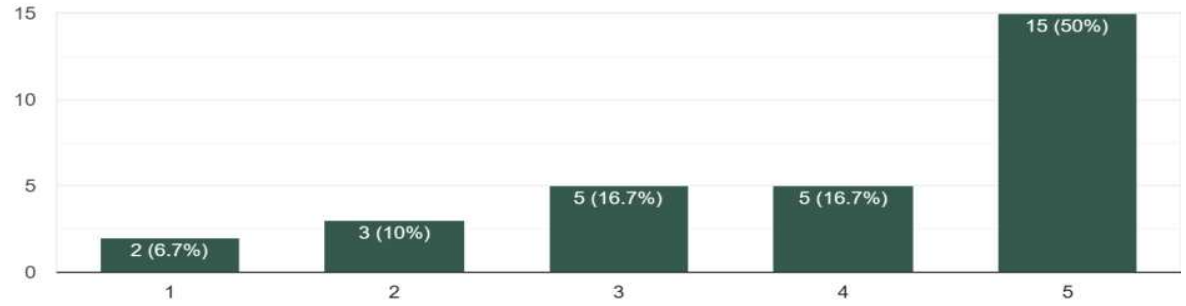
How important is a digital early warning system for your area? (আপনার এলাকায় ডিজিটাল আগাম সতর্কীকরণ ব্যবস্থা কতটা গুরুত্বপূর্ণ?)

30 responses



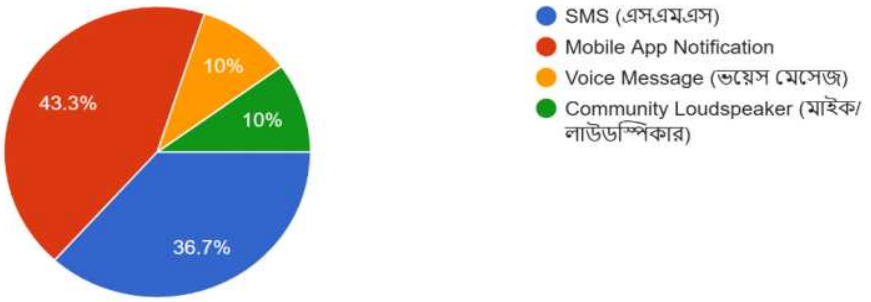
If a mobile app provided local flood alerts and relief information, how likely are you to use it? (যদি কোনো মোবাইল অ্যাপ স্থানীয় বন্যা সতর্কতা ও ত্রাণ তথ্য দেয়, আপনি কতটা ব্যবহার করবেন?)

30 responses



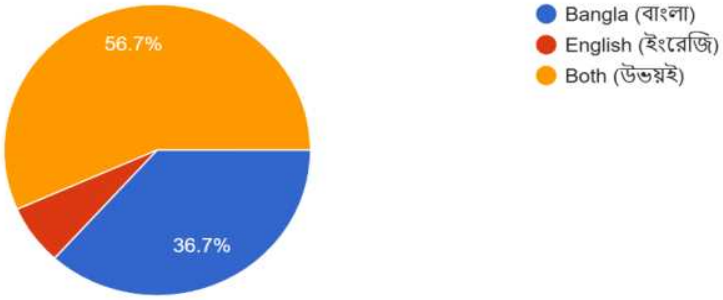
Which notification method would be easiest for you to receive alerts? (আপনার কাছে সতর্কতা পৌঁছানোর সবচেয়ে সহজ উপায় কোনটি?)

30 responses



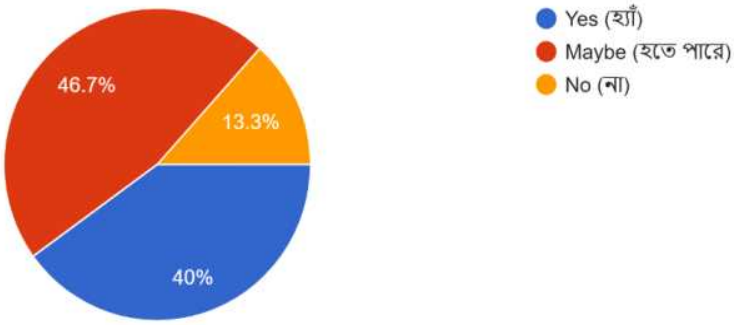
What language would you prefer for app notifications? (অ্যাপ নোটিফিকেশনের জন্য কোন ভাষা পছন্দ করবেন?)

30 responses



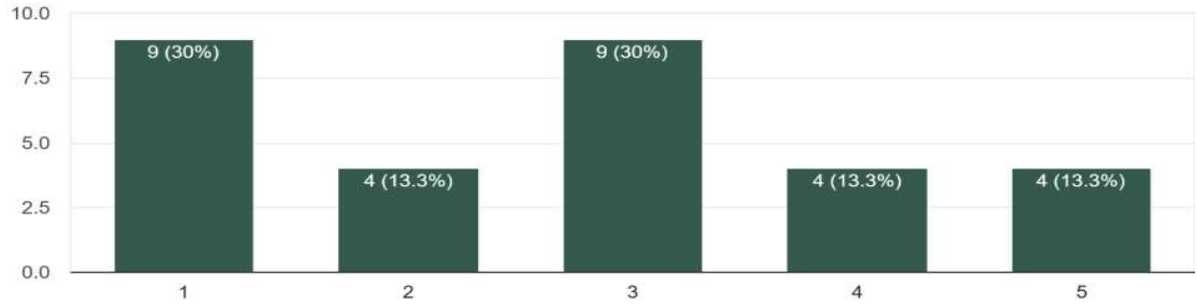
Do you think elderly or illiterate residents in your area could use a simple voice-alert system? (আপনার এলাকায় বয়স্ক বা নিরক্ষর মানুষ কি সহজ ভয়েস-অ্যালার্ট সিস্টেম ব্যবহার করতে পারবেন?)

30 responses



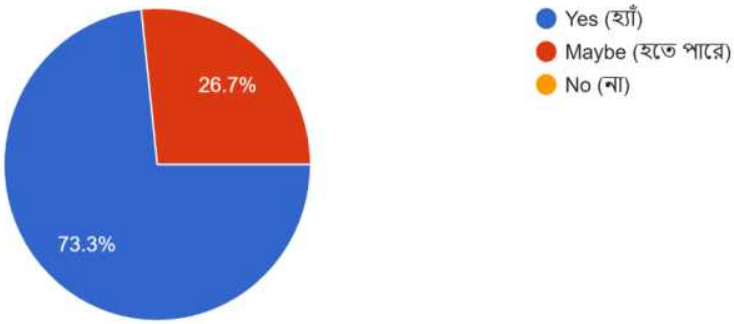
How satisfied are you with the local disaster management efforts? (স্থানীয় দুর্যোগ ব্যবস্থাপনা কার্যক্রমে আপনি কতটা সন্তুষ্ট?)

30 responses



Would you support using a digital platform for community management and flood response? (কমিউনিটি ম্যানেজমেন্ট ও বন্যা মোকাবিলায় ডিজিটাল প্ল্যাটফর্ম ব্যবহারের পক্ষে কি আপনি?)

30 responses



3.7 SWOT Analysis

	Helpful	Harmful
Internal	**Strengths:** • High tech readiness (88.9% smartphones) • Strong community support (74.1%) • Comprehensive feature set • Multi-channel approach	**Weaknesses:** • Communication dependency • Literacy barriers • Limited local management satisfaction (14.8%) • Implementation complexity
External	**Opportunities:** • High demand for early warning (51.9%) • Digital Bangladesh initiatives • NGO partnership potential • Scalability to other regions	**Threats:** • Corruption in existing systems • Digital exclusion of elderly (48.1%) • Infrastructure limitations • Funding sustainability

Table 3.2: SWOT Analysis for HaChar ManaRes

Chapter 4

System Design

4.1 Introduction to System Design

This chapter presents the architectural and detailed design of the HaChar ManaRes system. It illustrates how the system's components interact, the data flows, the structure of the database, and the overall system behavior. The design diagrams below will provide a visual representation of these aspects.

4.2 Use Case Diagram

The following diagram illustrates the comprehensive interaction model between various stakeholders and the Haor/Char Community Management & Disaster Response System. The system is designed to address the unique challenges of flood-prone Haor/Char regions by facilitating disaster preparedness, emergency response coordination, resource management, and sustainable community development through digital intervention.

4.2.1 Actors and Their Roles:

- **Resident** / **Community Member**: Primary end-user including farmers, fishermen, livestock owners, and general inhabitants who utilize services for safety, livelihood support, and community engagement.
- **Administrator**: System operator with full access to manage users, configurations, analytics, and overall system operations.
- **Authority** / **Government Body**: Local government representatives responsible for disaster management, policy enforcement, permit issuance, and official communications.
- **NGO** / **Non-Governmental Organization**: Aid organizations providing financial support, training, and community development programs including micro-loans and educational initiatives.
- **Supplier** / **Vendor**: Registered vendors supplying emergency relief materials, agricultural inputs, medical supplies, and other essential goods.
- **Volunteer**: Community members who register to assist during emergencies with tasks like rescue operations, distribution, and support services.

4.2.2 Core Use Case Categories:

Disaster Preparedness & Response

- Registration (includes Verification)
- Receive Early Warnings (includes Broadcast Alerts)
- Boat Transport Request (includes Validate Location)
- Volunteer Registration
- View Assigned Tasks
- Assign Volunteer Task

Livelihood Support & Management

- Crop Loss Reporting (includes Validate Report)
- Manage Livestock Health
- Issue Fishing Permit
- Request Micro Loan
- Order Inventory
- Supply Management

Community Services & Monitoring

- View Community Notices
- View Analytics Dashboard
- Track School Attendance

4.2.3 System Boundaries:

The system encompasses activities from individual registration and disaster alerting to complex resource management and analytical reporting. Key inclusion relationships are:

- Registration process **includes** Identity Verification
- Early Warning system **includes** Multi-channel Broadcast Alerts
- Transport Request **includes** Location Validation against flood maps
- Loss Reporting **includes** Report Authentication and Validation

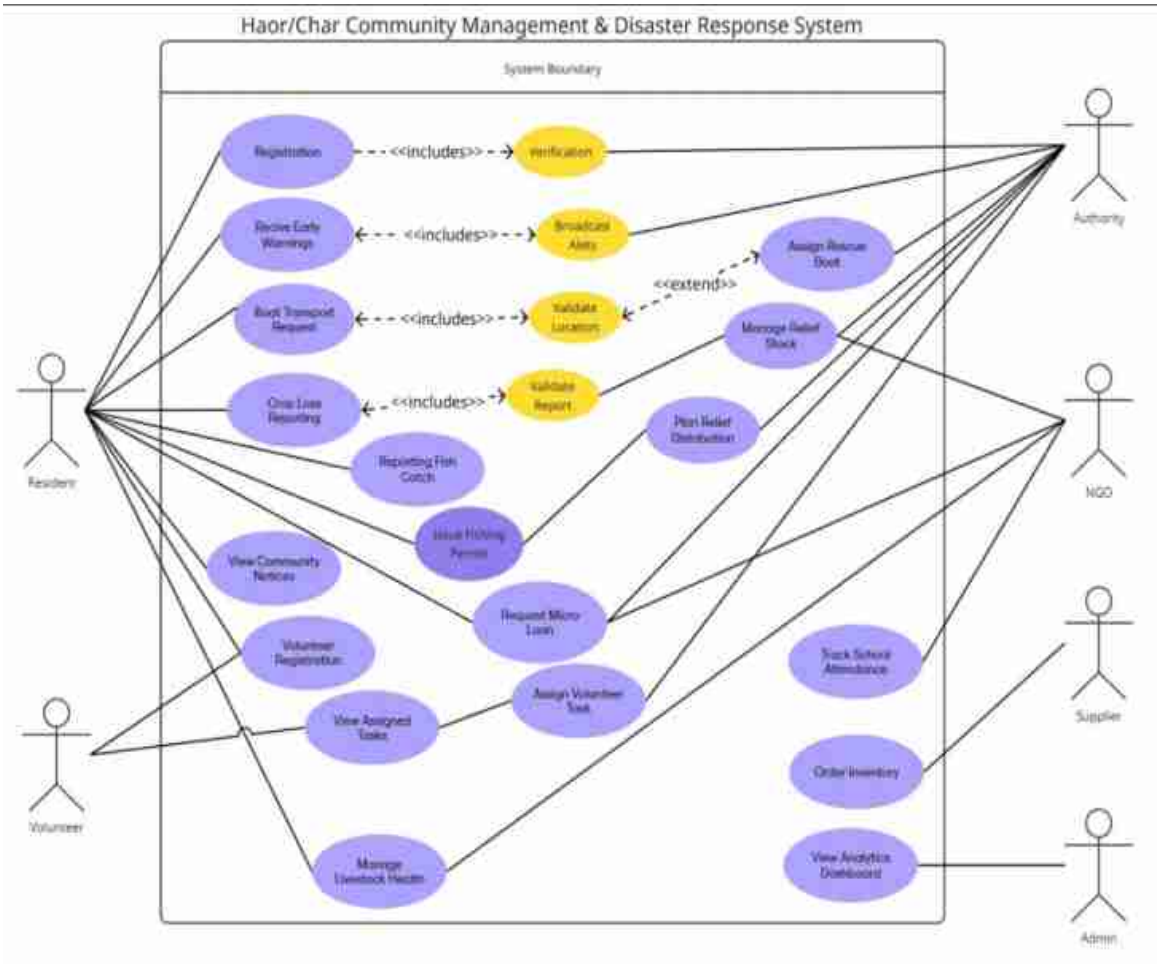


Figure 4.1: Use Case Diagram

4.3 Use Case Description

This section provides detailed use case descriptions for the following use case diagram.

4.3.1 Use Case 1: Registration

Registration	
Primary Actor	Resident / User
Secondary Actor(s)	System, Administrator, Authority
Description	Allows residents to create a verified account for accessing services.
Preconditions	User has application access and internet connection.
Postconditions	Profile created, verification pending, temporary credentials issued.
Main Success Scenario: 1. User enters required details → 2. System checks duplicates → 3. System saves profile → 4. User receives temporary credentials.	
Alternatives: Incomplete data → prompt for missing; Duplicate found → suggest login; Verification pending → limited access.	

4.3.2 Use Case 2: Receive Early Warnings

Receive Early Warnings	
Primary Actor	Resident
Secondary Actor(s)	System, Authority
Description	System sends real-time disaster alerts based on user location.
Preconditions	User verified with valid contact information.
Postconditions	User receives alert, system logs delivery status.
Main Success Scenario: 1. Authority creates alert → 2. System identifies affected users → 3. Broadcast via multiple channels → 4. User views alert → 5. System logs delivery.	
Alternatives: No connectivity → queue alert; Non-affected area → no alert; Creation fails → notify authority.	

4.3.3 Use Case 3: Boat Transport Request

Boat Transport Request	
Primary Actor	Resident
Secondary Actor(s)	System, Volunteer, Authority
Description	Request emergency or scheduled transport during floods.
Preconditions	User verified in flood-affected/serviceable area.
Postconditions	Request logged with tracking ID, queued for assignment.
Main Success Scenario: 1. User selects request → 2. GPS auto-fills location → 3. Specify purpose/urgency → 4. System validates location → 5. Request submitted with tracking ID.	
Alternatives: Location not serviceable → notify with options; Multiple requests → prioritize by urgency; GPS unavailable → manual entry.	

4.3.4 Use Case 4: Crop Loss Reporting

Crop Loss Reporting	
Primary Actor	Farmer / Resident
Secondary Actor(s)	System, Authority, NGO
Description	Report agricultural damage for assessment and compensation.
Preconditions	User registered as farmer with associated land records.
Postconditions	Report submitted for verification, added to assessment database.
Main Success Scenario: 1. Farmer enters loss details → 2. Uploads photo evidence → 3. System checks duplicates → 4. Report marked ”Under Verification”.	
Alternatives: Incomplete evidence → prompt more; Duplicate claim → guide to existing; Land unregistered → direct to registration.	

4.3.5 Use Case 5: View Community Notices

View Community Notices	
Primary Actor	Resident
Secondary Actor(s)	System, Authority, NGO
Description	Access public announcements and updates from authorities/NGOs.
Preconditions	None (public access).
Postconditions	User views relevant notices.
Main Success Scenario: 1. User navigates to notices → 2. System displays sorted notices → 3. User filters by category → 4. Views detailed information.	
Alternatives: No notices → display ”No current notices”; Network error → load cached notices.	

4.3.6 Use Case 6: Volunteer Registration

Volunteer Registration	
Primary Actor	Resident
Secondary Actor(s)	System, Administrator
Description	Register as volunteer with skills and availability for disaster response.
Preconditions	User has verified resident account.
Postconditions	Volunteer profile created and marked active.
Main Success Scenario: 1. User selects ”Become Volunteer” → 2. Specifies skills/availability → 3. Provides emergency contact → 4. System saves profile, notifies admin.	
Alternatives: Incomplete skills → suggest common skills; Underage → restrict registration.	

4.3.7 Use Case 7: View Assigned Tasks

View Assigned Tasks	
Primary Actor	Volunteer
Secondary Actor(s)	System, Administrator
Description	View, accept, decline, and update assigned task status.
Preconditions	User is approved and active volunteer.
Postconditions	Volunteer can interact with tasks and update completion status.
Main Success Scenario: 1. Volunteer navigates to ”My Tasks” → 2. System displays tasks → 3. Views task details → 4. Accepts/declines/completes task.	
Alternatives: No tasks → display message; Task expired → auto-mark expired, notify admin.	

4.3.8 Use Case 8: Manage Livestock Health

Manage Livestock Health	
Primary Actor	Livestock Owner / Farmer
Secondary Actor(s)	System, Veterinary Service, NGO
Description	Report sick animals, request veterinary visits, access health advice.
Preconditions	User registered as livestock owner.
Postconditions	Health report logged, veterinary notified if requested.
Main Success Scenario: 1. Owner adds animal details/symptoms → 2. System suggests possible diseases → 3. Owner requests veterinary visit → 4. System notifies veterinarians.	
Alternatives: Minor symptoms → provide self-care; Emergency → prioritize request, alert nearest vet.	

4.3.9 Use Case 9: Issue Fishing Permit

Issue Fishing Permit	
Primary Actor	Fisherman
Secondary Actor(s)	System, Authority
Description	Apply for or renew fishing permits digitally.
Preconditions	Fisherman has verified profile and valid ID.
Postconditions	Digital permit issued, stored, accessible to fisherman.
Main Success Scenario: 1. Fisherman applies with boat/gear details → 2. System checks restrictions → 3. Authority approves → 4. System generates digital permit with QR.	
Alternatives: Non-compliance → reject with reason; Pending verification → show ”Under Review”.	

4.3.10 Use Case 10: Request Micro Loan

Request Micro Loan	
Primary Actor	Resident
Secondary Actor(s)	System, NGO, Financial Authority
Description	Apply for small loans for recovery or livelihood support.
Preconditions	User verified, may have transaction history.
Postconditions	Application submitted for review.
Main Success Scenario: 1. User enters loan details → 2. System performs eligibility check → 3. Application forwarded to loan officer → 4. User receives reference number.	
Alternatives: Ineligible → notify with reasons; Missing docs → prompt for required documents.	

4.3.11 Use Case 11: Assign Volunteer Task

Assign Volunteer Task	
Primary Actor	Administrator / Authority
Secondary Actor(s)	System, Volunteer
Description	Create and assign tasks to volunteers based on skills and location.
Preconditions	Volunteers registered and available; task details defined.
Postconditions	Volunteer notified, task logged as ”Assigned”.
Main Success Scenario: 1. Admin creates task → 2. System suggests matching volunteers → 3. Admin assigns with deadline → 4. System notifies volunteer.	
Alternatives: No suitable volunteer → suggest broader criteria; Volunteer declines → notify admin for reassignment.	

4.3.12 Use Case 12: Order Inventory

Order Inventory	
Primary Actor	Administrator
Secondary Actor(s)	System, Supplier
Description	Place orders for emergency supplies from registered suppliers.
Preconditions	Supplier registered, items in catalog.
Postconditions	Order placed, supplier notified, inventory updated.
Main Success Scenario: 1. Admin browses catalog → 2. Selects items/quantities → 3. System calculates cost → 4. Admin confirms → 5. Supplier notified.	
Alternatives: Out of stock → suggest alternatives; Exceeds budget → alert, suggest adjustments.	

4.3.13 Use Case 13: View Analytics Dashboard

View Analytics Dashboard	
Primary Actor	Administrator
Secondary Actor(s)	System
Description	View comprehensive analytics on disaster impact, aid distribution, etc.
Preconditions	Administrator has appropriate permissions.
Postconditions	Administrator views/exports reports.
Main Success Scenario: 1. Admin navigates to dashboard → 2. System displays key metrics/charts → 3. Admin filters data → 4. Generates/downloads reports.	
Alternatives: No data → display message; Data delay → show progress indicator.	

4.3.14 Use Case 14: Track School Attendance

Track School Attendance	
Primary Actor	Authority / NGO
Secondary Actor(s)	System, Teacher, School
Description	Monitor daily student attendance to ensure safety and identify displacement.
Preconditions	Schools, teachers, students registered in system.
Postconditions	Attendance recorded, alerts triggered for prolonged absences.
Main Success Scenario: 1. Teacher marks attendance → 2. System records absences → 3. Authority views reports → 4. System flags prolonged absences.	
Alternatives: Teacher unable to mark → use previous day's estimate; Sudden drop → trigger immediate alert.	

4.3.15 Use Case 15: Supply Management

Supply Management	
Primary Actor	Supplier
Secondary Actor(s)	System, Administrator
Description	Receive orders, update stock, confirm deliveries through vendor portal.
Preconditions	Supplier has active vendor account.
Postconditions	Order status updated, inventory synchronized, admin notified.
Main Success Scenario: 1. Supplier views pending orders → 2. Updates order status → 3. System updates inventory → 4. Supplier generates/upload invoices.	
Alternatives: Order cancelled → mark with reason; Stock shortage → mark out of stock, suggest alternatives.	

4.4 Data Flow Diagram

4.4.1 Data Flow Diagram Description

The Data Flow Diagram (DFD) presented in Figure 4.2 models the information flow within the Haor/Char Community Management & Disaster Response System. This diagram captures the system's data processing architecture at a high level (Level 0/1 DFD), illustrating how data originates from various stakeholders, undergoes transformation through system processes, and is persistently stored for future reference and decision-making.

External Entities (Sources/Sinks)

The system interfaces with the following external entities:

- **Resident:** Primary data source for personal information, service requests, and feedback.
- **Administrator/Authority:** Source for system configuration, alert generation, permit approvals, and task assignments.
- **NGO:** Source for financial aid data, training schedules, and community program information.
- **Supplier:** Source for inventory catalogs, pricing, and order fulfillment status.
- **External Services:** Includes Weather Forecasting APIs (for early warnings), GIS Services (for location validation), and Communication Gateways (SMS/Email for alerts).

Core System Processes

The main data transformation processes include:

1. **P1: User Management Process** – Handles registration, verification, profile updates, and authentication.
2. **P2: Disaster Alerting Process** – Receives threat data, validates affected zones, formats alerts, and initiates multi-channel broadcasting.
3. **P3: Resource Coordination Process** – Manages requests for transport, supplies, and aid; validates eligibility; and assigns resources.
4. **P4: Livelihood Support Process** – Processes crop/livestock loss reports, loan applications, and permit requests.
5. **P5: Volunteer Coordination Process** – Matches volunteer skills with tasks, assigns work, and tracks completion.
6. **P6: Inventory Management Process** – Handles procurement, supplier coordination, and stock level maintenance.
7. **P7: Analytics & Reporting Process** – Aggregates data from all stores to generate dashboards, insights, and compliance reports.

Data Stores (Repositories)

Persistent data is maintained in the following repositories:

- **D1: Resident Registry** – Stores verified resident profiles, family trees, and contact details.

- **D2: Alert Log** – Archives all disseminated warnings, delivery receipts, and public responses.
- **D3: Service Request Repository** – Contains all active and historical requests for transport, aid, and support.
- **D4: Livelihood Asset Register** – Maintains records of land, livestock, fishing permits, and financial aid history.
- **D5: Volunteer Directory** – Holds volunteer profiles, skill sets, availability, and task history.
- **D6: Supply Chain Database** – Tracks suppliers, inventory items, purchase orders, and delivery logs.
- **D7: Analytics Warehouse** – Consolidated data store for reporting, trend analysis, and decision support.

Primary Data Flows

Critical data movements within the system include:

- **Registration Flow:** Personal Data → P1 → D1
- **Alert Flow:** Weather Data → P2 → D2 → Residents
- **Request Flow:** Service Need → P3 → D3 → P5/P6 (for fulfillment)
- **Verification Flow:** Claim Data → P4 → D4 (with cross-checks to D1)
- **Assignment Flow:** Task Details → P5 → D5 → Volunteers
- **Procurement Flow:** Order Request → P6 → D6 → Suppliers
- **Reporting Flow:** D1–D6 → P7 → D7 → Dashboards/Reports

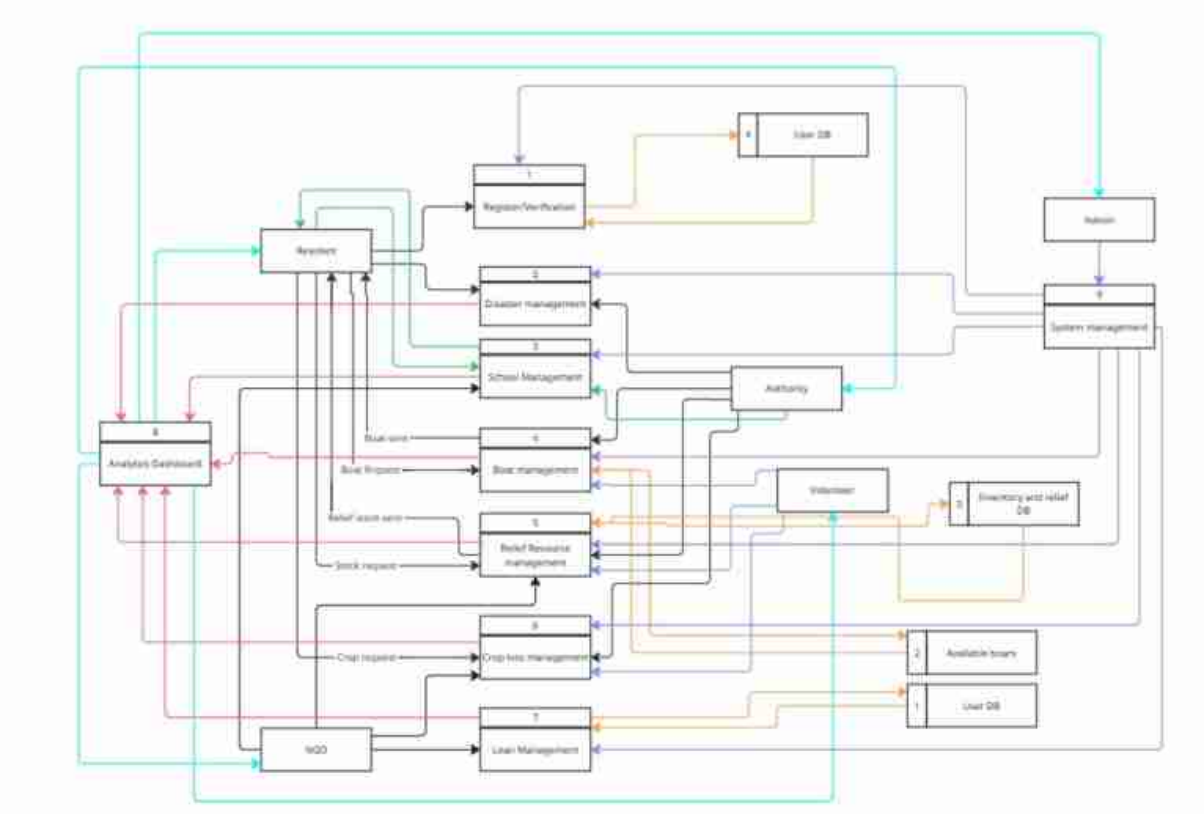


Figure 4.2: Data Flow Diagram

4.5 Class Diagram

System Class Structure provides a comprehensive view of the Haor/Char Community Management & Disaster Response System object-oriented design. The diagram illustrates:

- **Class Attributes and Methods:** Key properties and operations for each class.
- **Relationships:** Inheritance, associations, and dependencies between classes.
- **System Architecture:** Modular organization into cohesive packages representing disaster management, community support, and administrative functions.

The class diagram serves as the blueprint for system implementation and ensures maintainability, scalability, and adaptability to the evolving needs of flood-prone communities.

Core Class Categories:

- **Actor Classes:** Resident, Administrator, Authority, NGO, Supplier, Volunteer
- **Disaster Management Classes:** Alert, EmergencyRequest, TaskAssignment, ResourceAllocation
- **Livelihood Support Classes:** Farmer, Fisherman, CropLossReport, LoanApplication, FishingPermit
- **Resource Management Classes:** Inventory, SupplyOrder, DistributionRecord, BoatResource
- **Community Service Classes:** School, AttendanceRecord, HealthService, VeterinaryRecord
- **System Utility Classes:** NotificationService, ReportGenerator, AnalyticsEngine, Validator

Key Relationships:

- **Inheritance:** Volunteer extends Resident, Farmer extends Resident, etc.
- **Associations:** Resident submits EmergencyRequest, Administrator assigns Task to Volunteer
- **Aggregations:** Community contains multiple Residents, DisasterZone contains multiple Alerts
- **Dependencies:** AlertService depends on WeatherAPI, ReportGenerator depends on AnalyticsEngine

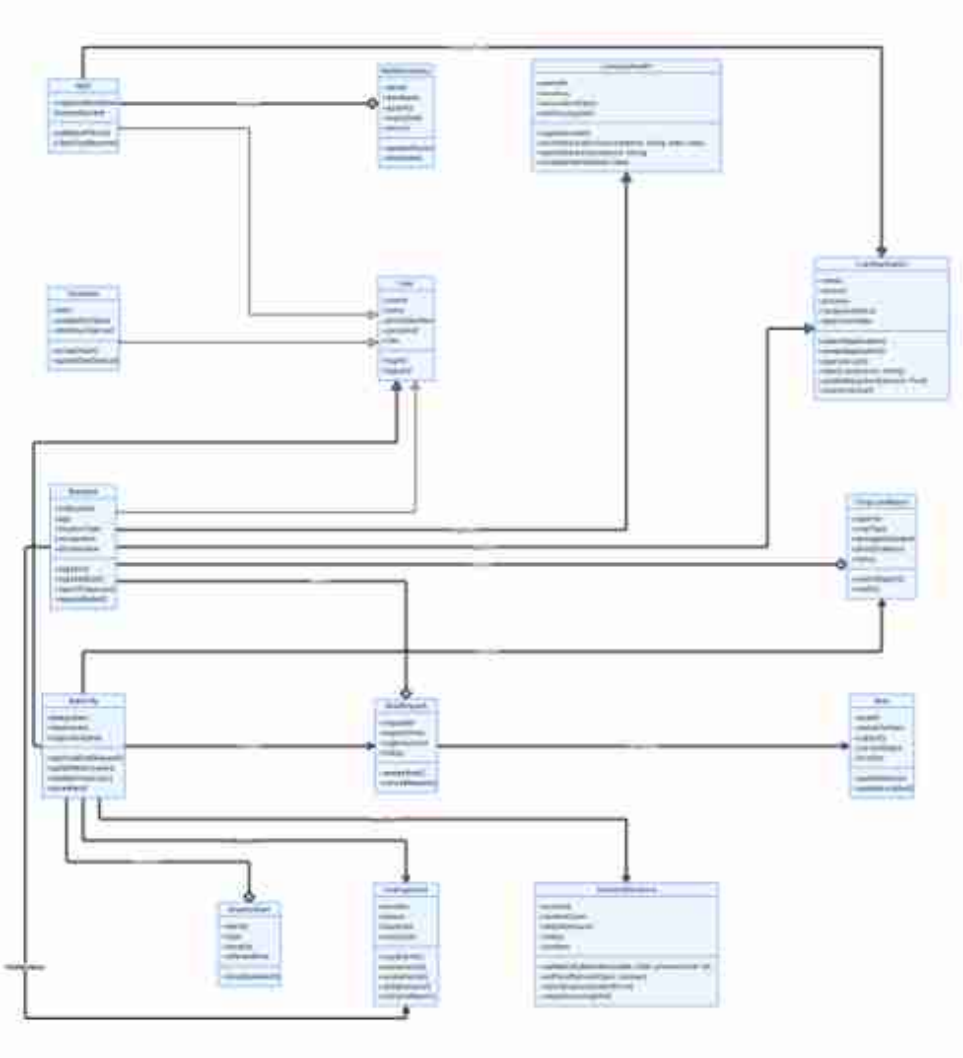


Figure 4.3: Class Diagram

4.6 Sequence Diagram

Interaction Flow illustrates the chronological message exchanges among actors and system components in the Haor/Char Community Management & Disaster Response System. The diagram captures three critical disaster response workflows:

Sequence 1: Emergency Transport Coordination

1. **Resident** sends `requestBoat(location, urgency)` to **System**
2. **System** invokes `Validate Location` service to confirm flood-affected zone
3. **System** creates **BoatRequest** object via `create(requestId, urgencyLevel)`
4. **System** calculates Estimated Time of Arrival (ETA)
5. **System** sends `Send Confirmation (ETA)` to **Resident**
6. **System** returns `Acknowledge Receipt` to complete the transaction

Sequence 2: Disaster Alert Dissemination

1. **System** performs `Calculate Risk Level` based on environmental sensors

2. **System** issues `Display High Risk Warning` to **Authority** dashboard
3. **Authority** triggers `issueAlert(type, severity, area)` to **System**
4. **System** creates **DisasterAlert** object via `create(alertId, type, area)`
5. **System** executes `broadcastAlert()` to notification channels
6. **System** sends `Send SMS/App Notification` to all **Residents** in affected area

Sequence 3: Agricultural Compensation Processing

1. **Farmer** submits `reportCropLoss(cropType, estimatedDamage)` to **System**
2. **Farmer** provides `Upload Photo Evidence` for verification
3. **Farmer** calls `submitReport()` to finalize submission
4. **System** creates **CropLossReport** object and returns `Confirm Submission`
5. **Authority** accesses `viewReports()` to see pending claims
6. **System** provides `List Pending Reports`
7. **Authority** performs `validateCropLoss(reportId)` for assessment
8. **Authority** updates status via `verify(status="Approved")`
9. **System** sends `Notify Approval/Subsidy` to the **Farmer**

Key Design Patterns in Message Flow:

- **Command Pattern:** Encapsulates requests as objects (BoatRequest, DisasterAlert)
- **Observer Pattern:** Notifications broadcast to multiple residents simultaneously
- **Facade Pattern:** Simplified interfaces for complex validation processes
- **State Pattern:** Status transitions in CropLossReport (Pending → Validated → Approved)

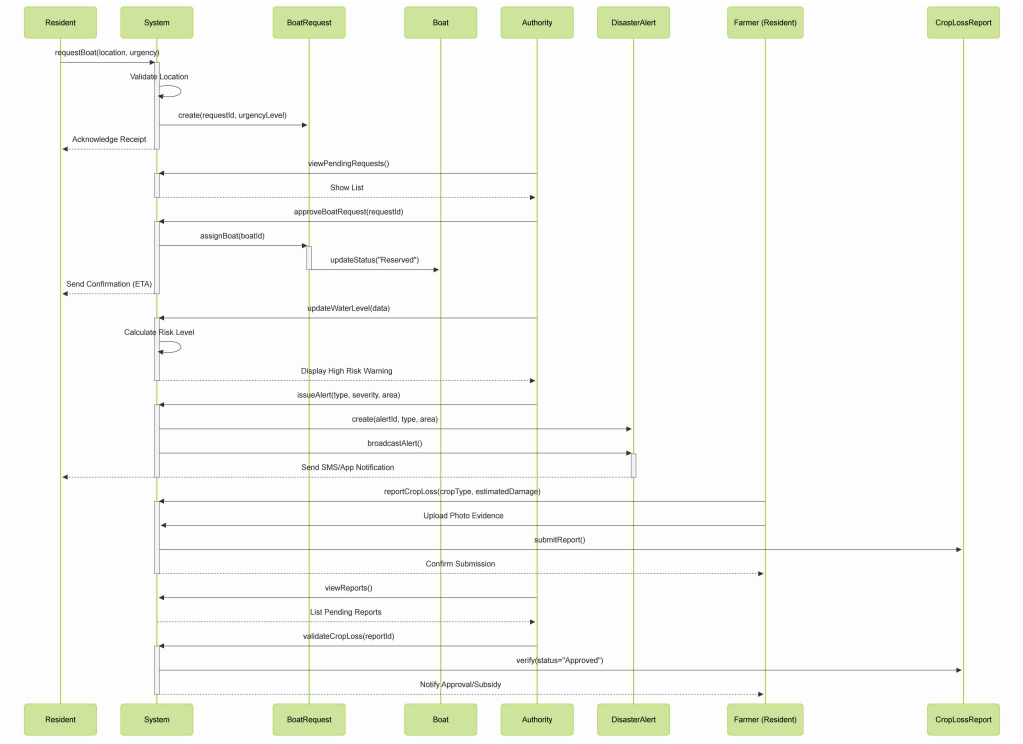


Figure 4.4: Sequence Diagram

Chapter 5

UI & UX Design

5.1 Prototyping Tools Used

- **Figma:** Primary tool for high-fidelity prototype design
- **Miro:** Collaborative brainstorming and user flow mapping
- **Google Gemini:** Generate system graphs and chart

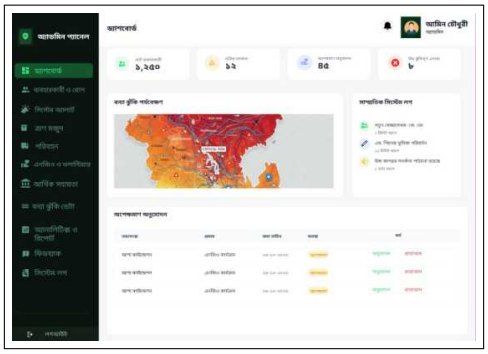
5.2 UI Design Principles

1. **Accessibility First:** Design for users with varying literacy and tech skills
2. **Context-Aware:** Interface adapts to emergency vs normal situations
3. **Minimalist Design:** Simple, clear interfaces for high-stress situations
4. **Localization:** Full Bangla support with cultural appropriateness
5. **Offline Capability:** Core functions available without internet
6. **Progressive Disclosure:** Advanced features revealed as needed

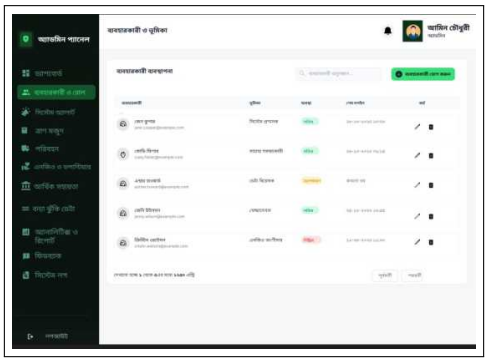
5.3 Figma Prototype Gallery

This section provides a dedicated space for the full high-fidelity Figma prototype (**25 screens**) of the Haor/Char Community Management & Disaster Response System, showcasing the user interface design across all functional modules.

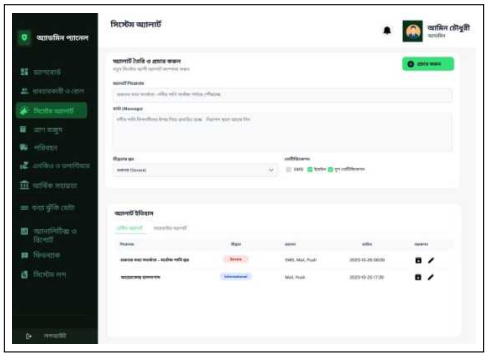
Group 1: Admin Dashboard



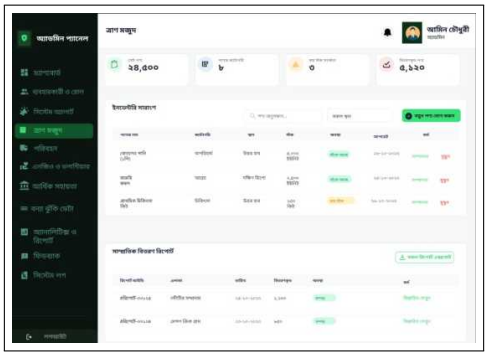
(a) Screen 01 — Admin Dashboard



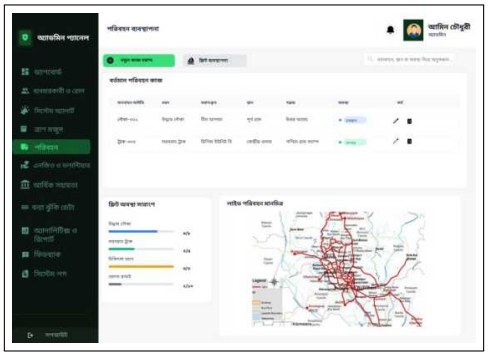
(b) Screen 02 — User Logs



(c) Screen 03 — System Alerts



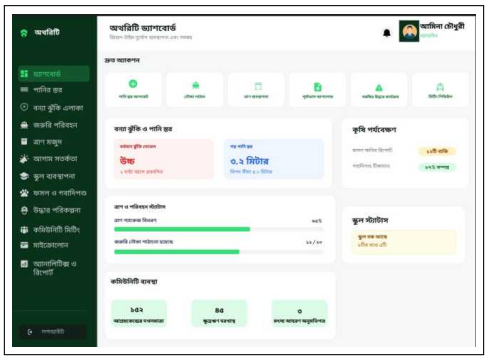
(d) Screen 04 — Relief Management



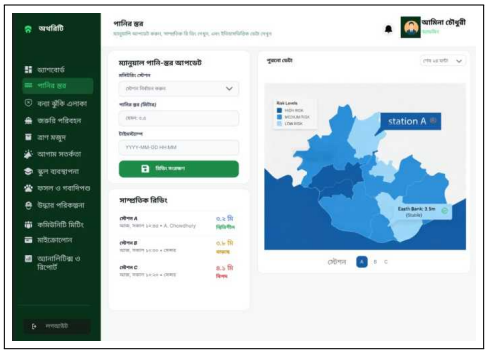
(e) Screen 05 — Transport Management

Figure 5.1: Admin's Screens

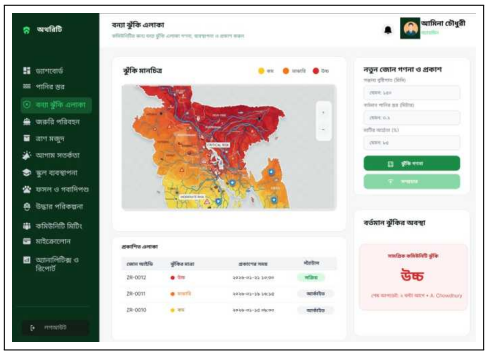
Group 2: Authority Dashboard



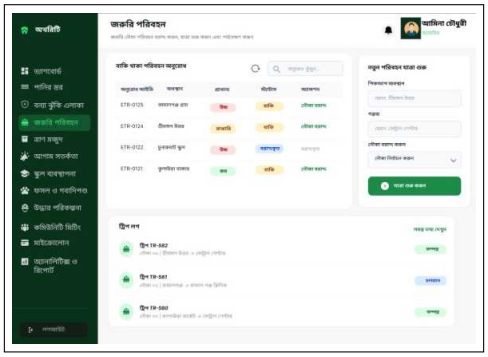
(a) Screen 06 — Authority Dashboard



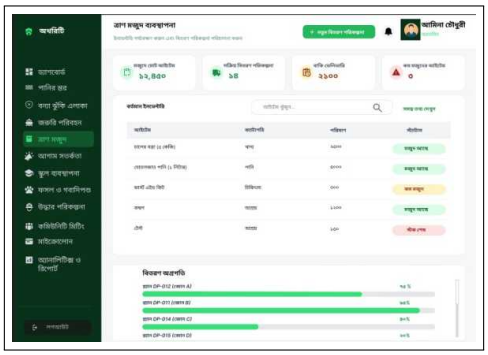
(b) Screen 07 — Water Level Update



(c) Screen 08 — Flood Risk Management



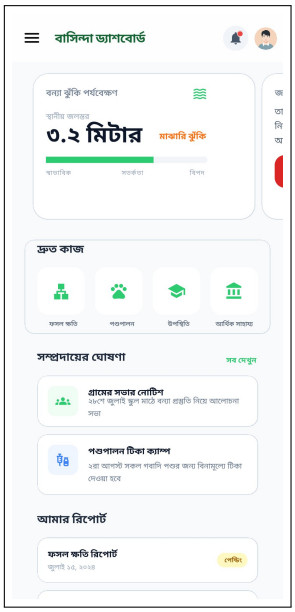
(d) Screen 09 — Emergency Evacuation



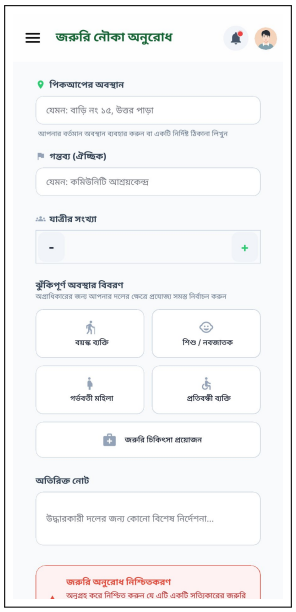
(e) Screen 10 — Relief Stock Management

Figure 5.2: Authority's Screens

Group 3: Resident Dashboard



(a) Screen 11 — Resident Dashboard



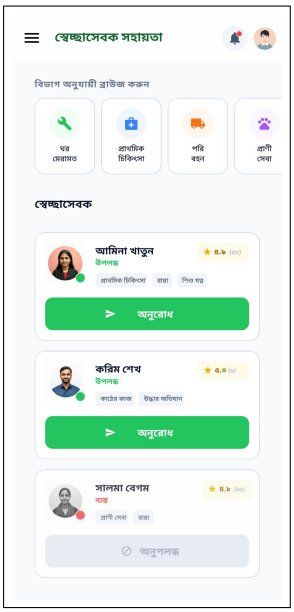
(b) Screen 12 — Emergency Boat Request



(c) Screen 13 — Flood Risk Monitoring



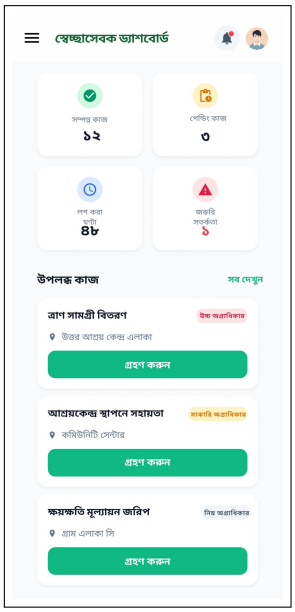
(d) Screen 14 — School Monitoring



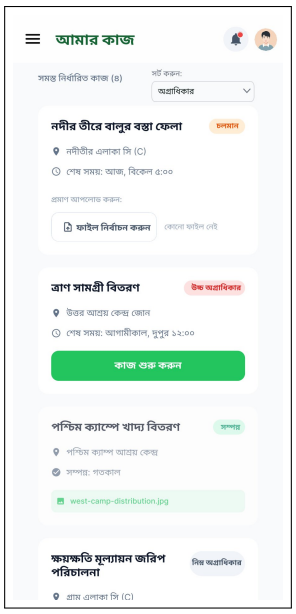
(e) Screen 15 — Volunteer Support

Figure 5.3: Residen's Screens

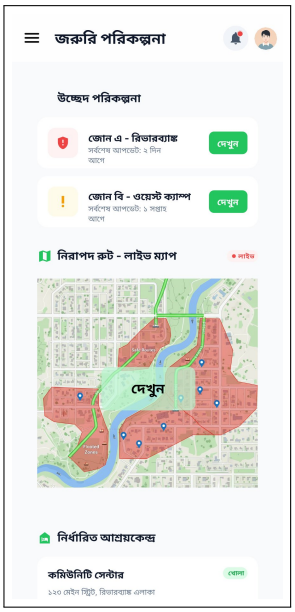
Group 4: Volunteer Dashboard



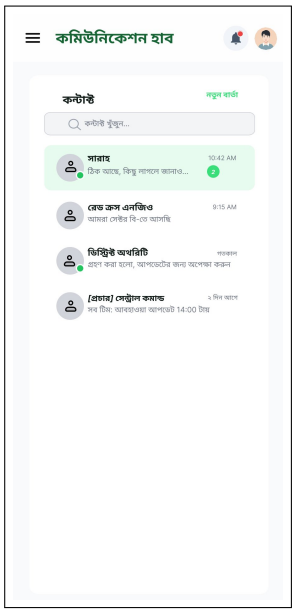
(a) Screen 16 — Volunteer Dashboard



(b) Screen 17 — My Tasks



(c) Screen 18 — Emergency Evacuation Execution



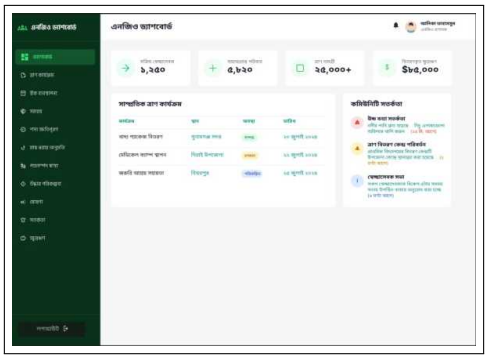
(d) Screen 19 — Communication Hub



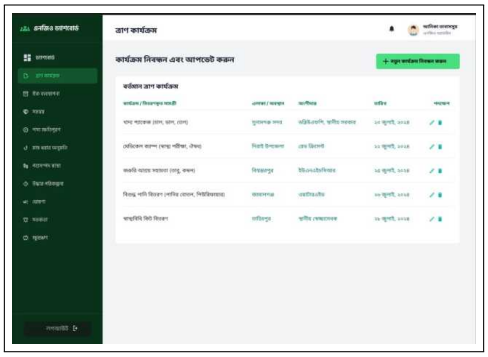
(e) Screen 20 — Feedback & Report

Figure 5.4: Volunteer's Screens

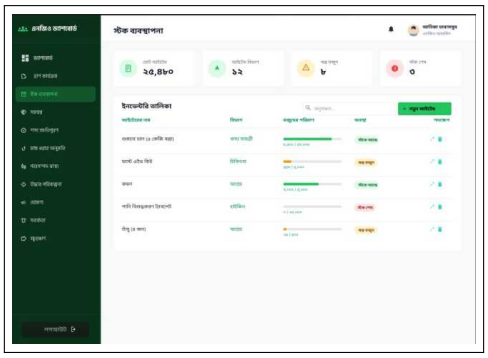
Group 5: NGO Dashboard



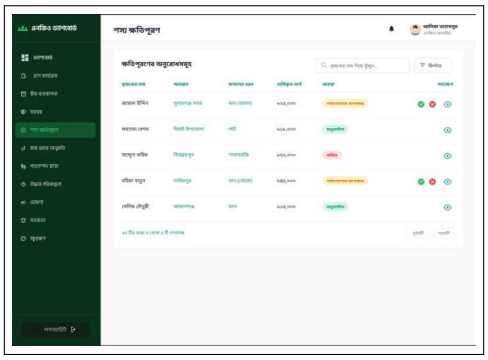
(a) Screen 21 — NGO Dashboard



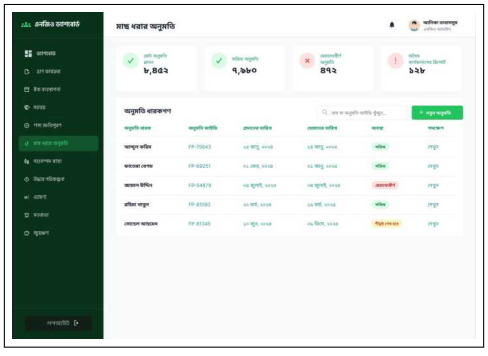
(b) Screen 22 — Relief Activity Management



(c) Screen 23 — Inventory Management



(d) Screen 24 — Crop Compensation



(e) Screen 25 — Fishing Permit

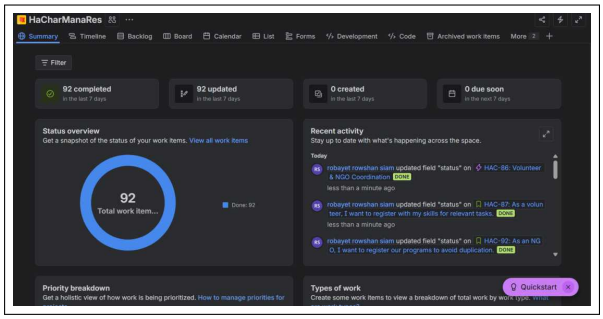
Figure 5.5: NGO's Screens

5.4 Jira Implementation & Project Tracking

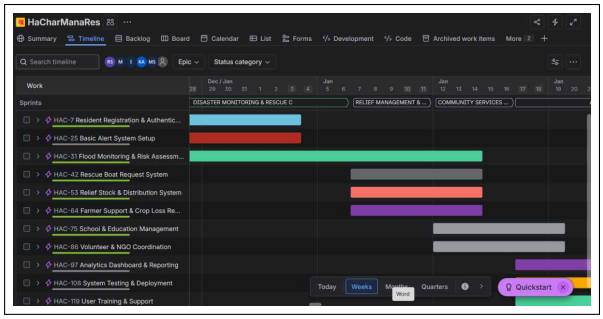
This section documents how Jira is used to plan, track, and report progress for the HaChar ManaRes project. The Jira board maps each UI/feature item from the Figma

prototype to an Epic/Story/Task, ensuring end-to-end traceability.

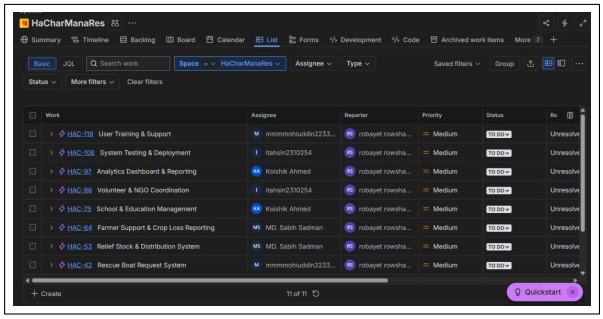
Project Management Screenshots (Jira/Agile Tools)



(a) Haor/Char System Project Dashboard



(b) Product Backlog & Sprint Planning



(c) Sprint Board (To Do → In Progress → Done)

Figure 5.6: Key Jira Screens

Chapter 6

Features

6.1 Functional Features

1. **Resident Management:**
   - Profile creation with NID verification
   - Vulnerability tagging (pregnant, elderly, disabled)
   - Occupation categorization (farmer, fisherman, laborer)
   - Location mapping to Haor/Char areas
2. **Flood Risk Monitoring:**
   - Daily water level updates by authorities
   - Automated risk level generation (Low/Medium/High)
   - Trend-based flood prediction
   - Color-coded risk visualization maps
3. **Boat Transport System:**
   - Rescue and transport request interface
   - Nearest available boat assignment
   - Estimated arrival time calculation
   - Trip logging and history maintenance
4. **Relief Management:**
   - Warehouse stock tracking (rice, lentils, medicines)
   - Distribution planning and assignment
   - Transparency reports generation
   - Donation tracking and allocation
5. **Early Warning System:**
   - Multi-channel alerts (SMS, app, voice)
   - Weather forecast integration
   - Cyclone and flash flood warnings
   - Audio alerts for illiterate users
6. **Agricultural Support:**
   - Crop loss reporting with photo verification
   - Government subsidy tracking
   - Seasonal crop recommendations
   - Farmer group coordination
7. **Community Support:**

- Digital fishing permit issuance
- Illegal fishing activity reporting
- Fishing ban period notifications
- Catch reporting and analytics

8. **Livestock Management:**
   - Animal registration and health logs
   - Vaccination schedule tracking
   - Veterinary visit scheduling
   - Disease outbreak alerts
9. **Evacuation Planning:**
   - Shelter identification and mapping
   - Live occupancy monitoring
   - Vulnerable resident prioritization
   - Rescue transport allocation
10. **NGO Coordination:**
    - Activity registration and tracking
    - Resource duplication prevention
    - Joint planning and coordination
    - Impact assessment reporting

6.2 Non-functional Features

1. **Performance:**
   - Response time ¡ 3 seconds for critical functions
   - Support for 10,000+ concurrent users during emergencies
   - 99.5% system availability during disaster periods
   - Efficient data synchronization in low-bandwidth conditions
2. **Security:**
   - Role-based access control (RBAC)
   - End-to-end encryption for sensitive data
   - Regular security audits and vulnerability assessments
   - GDPR-compliant data protection
3. **Usability:**
   - Intuitive interface with Bangla language support
   - Accessibility features for disabled users
   - Voice-based interaction for illiterate users
   - Context-sensitive help and tutorials
4. **Reliability:**
   - 24/7 operational capability
   - Automated backup and disaster recovery
   - Redundant systems for critical functions
   - Regular maintenance and updates
5. **Scalability:**
   - Modular architecture for easy feature addition
   - Cloud-based infrastructure for elastic scaling
   - Support for multiple regions and communities
   - API-based integration capabilities

Chapter 7

Implementation Strategy & Future Roadmap

7.1 Technology Stack

Component	Technology	Rationale
Frontend	React.js (PWA)	Cross-platform, offline capability
Mobile App	React Native	Single codebase for iOS/Android
Backend	Node.js with Express	High performance, real-time capabilities
Database	MongoDB + PostgreSQL	Flexible schema + relational integrity
Real-time	Socket.io	Instant notifications and updates
SMS Gateway	Twilio API	Reliable SMS delivery
Mapping	Leaflet.js + OpenStreetMap	Lightweight, open-source mapping
Hosting	AWS/Azure Cloud	Scalability and reliability

Table 7.1: Technology Stack for HaChar ManaRes

7.2 Deployment Strategy

1. **Pilot Phase:**
   - 3 selected Haor communities (5,000 residents)
   - 6-month testing and refinement period
   - Community training and onboarding
   - Feedback collection and system optimization
2. **Expansion Phase:**
   - Scale to 20+ communities in 2 years
   - Integration with national disaster management systems
   - Mobile network operator partnerships
   - Local government adoption and ownership

3. **Sustainability Phase:**
   - Community-led maintenance and support
   - Revenue model for long-term sustainability
   - Continuous feature development
   - Regional replication potential

7.3 Future Enhancements

1. **AI/ML Integration:**
   - Predictive analytics for flood forecasting
   - Image recognition for damage assessment
   - Chatbots for user support
   - Anomaly detection in resource distribution
2. **Advanced Analytics:**
   - Predictive modeling for disaster impact
   - Social network analysis for community resilience
   - Economic impact assessment tools
   - Climate change adaptation planning

7.4 Conclusion

The HaChar ManaRes System represents a comprehensive digital solution for addressing the unique challenges faced by Haor and Char communities in Bangladesh. By integrating disaster response with community management, the system provides a holistic approach to building resilience and improving livelihoods. The platform's design, informed by extensive community research and stakeholder engagement, ensures relevance, usability, and sustainability.

The system's success will be measured by:

- Reduced response time during disasters
- Improved transparency in resource distribution
- Enhanced community participation in disaster management
- Increased agricultural and fishing productivity
- Strengthened coordination among all stakeholders

With its scalable architecture and adaptable design, HaChar ManaRes has the potential to transform disaster management not only in Bangladesh but also in similar flood-prone regions worldwide.